

Business Requirement Document (BRD)

AcademiCore Instrument Allocation System

Domain: Student Management System
Project: AcademiCore Instrument Allocation System (IAS)
Prepared By: Project Team

1. Document Control

Version	Date	Author	Description
1.0	March 2026	Project Team	Initial BRD Draft

2. Executive Summary

The Academic Core Instrument Allocation System (IAS) is a comprehensive web-based platform specifically designed to manage laboratory instrument reservations for university students and staff. Currently, the instrument allocation process across departments such as ICT, IAT, ET, and AT is handled manually through paper logs or spreadsheets. This traditional approach introduces several critical challenges, including booking conflicts, a lack of transparency, inefficient tracking, and potential misuse of institutional resources.

The proposed automated system aims to streamline these operations by introducing a secure, centralized portal that enables users to seamlessly perform the following actions:

- Browse and check real-time instrument availability
- Reserve instruments online for designated academic time slots
- Track, manage, and cancel upcoming or past bookings
- Maintain structured operational oversight for faculty and administration

3. Business Objectives

The primary objectives and goals of implementing the AcademiCore Instrument Allocation System include:

- **Process Automation:** Drastically reduce reliance on manual, paper-based instrument allocation workflows and minimize human error.
 - **Conflict Prevention:** Completely eliminate double-booking of laboratory equipment through automated scheduling validation.
 - **Real-Time Visibility:** Provide users and administrators with instant updates regarding instrument status, current locations, and availability timelines.
 - **Operational Efficiency:** Empower laboratory staff and technical officers to easily manage inventory, approve or reject requests, and monitor resource usage.
 - **Traceability and Accountability:** Maintain comprehensive usage histories and logs to ensure fair, transparent, and responsible utilization of institutional equipment across all departments.
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4. Project Scope

4.1 In Scope

The system encompasses the following core features and modules:

- **User Authentication & Authorization:** Secure login utilizing university credentials with role-based access control (Student, Lab Staff, Administrator).
- **Instrument Catalog & Browsing:** Detailed listing of laboratory equipment with multi-parameter filtering options (by category, department, and availability status).
- **Online Reservation System:** Ability for verified students to select specific dates and time slots to reserve required laboratory instruments.
- **Booking Management & Approval Workflow:** Workflow tools allowing staff to review, approve, or reject reservation requests based on departmental guidelines.
- **Cancellation & Status Updates:** Self-service booking cancellation for students and real-time status management (e.g., "Available", "In Use", "Under Maintenance") for technical officers.
- **Automated Notifications:** Email notification dispatch for booking confirmations, modification updates, rejections, and cancellation alerts.
- **Reporting & Administrative Dashboard:** Full administrative control panel for user role management, inventory database controls, and generation of department-level utilization reports.

4.2 Out of Scope

The following components and integrations are excluded from the initial phase of the project:

- **Online Payment Integration:** Processing financial penalties, deposits, or rental fees via online gateways.

- **External Third-Party Rentals:** Integration or sharing of data with external commercial equipment rental enterprises.
 - **Native Mobile Applications:** Dedicated standalone mobile apps for iOS and Android platforms (the system will instead rely on a fully responsive web application layout accessible via mobile browsers).
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5. Stakeholder Analysis

Stakeholder	Primary Role & Responsibilities
Students	Browse the equipment catalog, check availability, submit reservation requests, and manage personal bookings.
Laboratory Staff / Technical Officers	Monitor incoming booking requests, approve/reject reservations, update equipment statuses, and manage lab logs.
System Administrators	Oversee high-level system configurations, manage user accounts, assign role permissions, and generate analytics reports.
University Management	Monitor overall resource distribution, evaluate lab utilization metrics, and enforce institutional policies.
Development Team	Design, develop, test, and deploy the web application while ensuring compliance with technical specifications.

6. Detailed Business Requirements

- **BR-01:** The system must provide a secure portal allowing authenticated students to view comprehensive lists of available instruments filtered by department and category.
- **BR-02:** The system must enable eligible students to submit online requests to reserve laboratory instruments for designated academic dates and time slots.
- **BR-03:** The system must empower designated laboratory staff to review pending requests and approve or reject them according to departmental policies.
- **BR-04:** The system must automatically track and store complete instrument usage histories and booking audit logs for accountability.

- **BR-05:** The system must grant administrators full authority to add, edit, archive, or delete instrument profiles and manage user permissions.
 - **BR-06:** The system must provide administrative tools to compile and generate analytical utilization reports across various departments.
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7. Functional Requirements

ID	Requirement Name	Priority & Detailed Description
FR-01	User Registration & Login	Users must securely log into the system using verified university email credentials, routing them to role-specific dashboards. (High Priority)
FR-02	Search & View Instruments	Students must be able to search the inventory and apply filters based on department, category, and real-time availability. (High Priority)
FR-03	Reserve Instruments	Students must be able to select specific dates and time intervals to reserve available items, subject to system validation checks. (High Priority)
FR-04	Booking Approval Workflow	Staff members must be able to view pending reservation queues and approve or reject them based on internal rules. (Medium Priority)
FR-05	Update Instrument Status	Staff must be able to update individual instrument conditions (e.g., available, in use, under maintenance, damaged). (High Priority)

ID	Requirement Name	Priority & Detailed Description
FR-06	Manage Inventory	Administrators must have full capability to add new equipment records, modify metadata, or archive broken items. (High Priority)
FR-07	Generate System Reports	Administrators must be able to generate structured usage statistics and export allocation records per department. (Low Priority)

8. Non-Functional Requirements

Category	Requirement Description
Performance	The system must support at least 100 concurrent users under normal operating loads and deliver page response times within 2 seconds.
Security	All user credentials and personal data must be protected using secure hashing, and all data in transit must be encrypted via enforced HTTPS/SSL certificates.
Usability	The user interface must maintain a clean, intuitive layout with a responsive design framework (such as Bootstrap) ensuring seamless accessibility across desktop, tablet, and mobile browsers.
Availability	The system must aim for a minimum of 99.5% uptime to ensure reliable around-the-clock availability for academic planning.

Category	Requirement Description
Reliability & Data Integrity	The underlying relational database must enforce strict normalization rules and validation logic to prevent overlapping bookings, complemented by automated daily data backups.
Maintainability	The backend architecture must be built using a modular design approach (e.g., Laravel MVC framework) to simplify future updates, scaling, and debugging.

9. Business Process Overview

- **Current Manual Process:**
 1. Student physically or informally requests an instrument.
 2. Staff manually checks physical logs or spreadsheets for availability.
 3. Staff writes down the reservation on paper or a local file.
 4. Instrument is handed over. (*Bottlenecks: Highly time-consuming, frequent scheduling conflicts, zero real-time visibility*).
 - **Proposed Automated Process:**
 1. Student logs into the web platform using university credentials.
 2. Student filters the digital catalog and reviews real-time equipment availability.
 3. Student submits an online reservation request for a specific time slot.
 4. System performs automated availability checks; staff reviews and approves the request if necessary.
 5. System records the transaction, updates equipment status, and dispatches an automated confirmation email.
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10. Assumptions & Constraints

- **Assumptions:**
 - All prospective users (students and staff) possess reliable internet-connected devices and active university email addresses.
 - Respective academic departments will proactively maintain and update their equipment inventory records.
 - Standardized booking rules (maximum duration limits, cancellation policies) are pre-defined prior to deployment.

- **Constraints:**
 - Limited project development time frame and academic milestone deadlines.
 - Implementation must operate within existing university infrastructure and IT security compliance frameworks.
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11. Success Metrics (KPIs)

The success and impact of the AcademiCore Instrument Allocation System will be evaluated against the following metrics:

- An estimated 80% reduction in administrative effort and manual booking paperwork.
- Complete elimination of scheduling conflicts and double-bookings.
- Noticeable increase in laboratory equipment utilization efficiency and transparency across participating departments.
- Enhanced user satisfaction scores reported by students and technical staff regarding booking turnaround times.